# BRAINSTORM GROUP (BSG)

# MAT111
# (Pure Mathematics)
# E-Test Questions [2021-22]

**Federal University of Technology Minna**

*09068836600, 09068836661* *brainstormgroupfutmx001@gmail.com*

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# BRAINSTORM GROUP (BSG)

## MAT111 E-TEST QUESTIONS (2021/22)

1. *Given* $Z = 4 - 2\sqrt{2}i$, *find* $Z\bar{Z}$

   A. 20 B. 22

   C. 24 D. 21

   ***[2021/22 e-Test]***

2. *Given the universal set* $U = \{1, 2, 3, 4, 5, 6\}$ *where* $X = \{2, 4, 6\}$ *and* $Y = \{1, 2, 6\}$, *find* $(X \cap Y)^c$.

   A. $\{2, 4, 5, 6\}$ B. $\{1, 3, 4, 5\}$

   C. $\{3, 5, 6, 7\}$ D. $\{1, 4, 5\}$

   E. $\{3, 4, 5\}$

   ***[2021/22 e-Test]***

3. *Find the imaginary part of* $\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)^2$

   A. $-\frac{1}{2}\sqrt{3}i$ B. $\frac{1}{2}\sqrt{3}i$

   C. $-\frac{1}{4}\sqrt{3}i$ D. $\frac{1}{4}\sqrt{3}i$

   ***[2021/22 e-Test]***

4. *Resolve* $\frac{3x+2}{(x+2)(x-5)}$ *into partial fraction*

   A. $\frac{1}{7}\left(\frac{17}{(x+2)}+\frac{4}{(x-5)}\right)$

   B. $\frac{1}{7}\left(\frac{4}{(x+2)}-\frac{17}{(x-5)}\right)$

   C. $\frac{1}{7}\left(\frac{4}{(x+2)}+\frac{17}{(x-5)}\right)$

   D. $\frac{1}{7}\left(\frac{17}{(x+2)}-\frac{4}{(x-5)}\right)$

   ***[2021/22 e-Test]***

5. *Find* $\frac{Z_1}{Z_2}$ *if* $Z_1 = 2 - 4i$ *and* $Z_2 = 6 + 7i$

   A. $\left(\frac{16}{85}+i\frac{38}{85}\right)$

   B. $\left(\frac{15}{85}-i\frac{38}{85}\right)$

   C. $-\left(\frac{16}{85}-i\frac{38}{85}\right)$

   D. $-\left(\frac{16}{85}+i\frac{38}{85}\right)$

   ***[2021/22 e-Test]***

6. In an examination, 18 students passed MAT111, 17 students passed PHY113, 11 students passed both subject. Find the number of students that passed MAT111 only.

   A. 5 B. 6

   C. 7 D. 8

   ***[2021/22 e-Test]***

7. *If* $Z_1 = 1 - 3i$ *and* $Z_2 = -2 + 5i$, *determine* $\frac{Z_1 Z_2}{Z_1+Z_2}$ *in the form* $a + ib$

   A. $-\frac{9}{5}+\frac{37}{5}i$ B. $\frac{9}{5}-\frac{37}{5}i$

   C. $-\frac{12}{25}-\frac{13}{25}i$ D. $\frac{9}{25}-\frac{37}{25}i$

   ***[2021/22 e-Test]***

8. *Find the real part of* $\frac{2+3i}{3+2i}$

   A. $\frac{12}{13}$ B. $-\frac{5}{13}$

   C. $\frac{5}{13}$ D. $-\frac{12}{13}$

   ***[2021/22 e-Test]***

9. Given that $\frac{3x^2-3x-2}{(x-1)(x+1)} = A + \frac{Bx+C}{(x-1)(x+1)}$, what is the value of A, B and C.

   A. $(3, 3, -2)$ B. $(2, 3, 1)$

   C. $(3, -3, -1)$ D. $(3, 3, 1)$

   ***[2021/22 e-Test]***

10. What is the partial fraction equivalent of $\frac{1}{(ax+b)^2}$

    A. $\frac{(Ax+B)^2}{ax+b}$ B. $\frac{Ax+B}{ax+b}$

    C. $\frac{A}{(ax+b)^2}+\frac{B}{(ax+b)^3}$

    D. $\frac{A}{ax+b}+\frac{B}{(ax+b)^2}$

    ***[2021/22 e-Test]***

11. *What is the value of* $\boldsymbol{a}$ *given* $a + ib = \frac{1-i}{2+i}$

    A. $-\frac{1}{5}$ B. $\frac{3}{5}$

    C. $-\frac{3}{5}$ D. $\frac{1}{5}$

    ***[2021/22 e-Test]***

12. What is the partial fraction equivalent of the rational function $\frac{1}{ax^2+bx+c}$

    A. $\frac{Ax^2+B}{ax^2+bx+c}$ B. $\frac{(Ax+B)(C+Dx)}{ax^2+bx+c}$

    C. $\frac{A}{ax^2+bx+c}+\frac{B}{ax+b}$ D. $\frac{Ax+B}{ax^2+bx+c}$

    ***[2021/22 e-Test]***

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13. If $\frac{5+x}{(2x^2-3)(x-1)} = \frac{Ax+B}{2x^2-3} + \frac{C}{x-1}$, *find the value of* $C$

A. 6 B. $-6$
C. $-5$ D. 5

***[2021/22 e-Test]***

14. *If* $\frac{x^2-4x+5}{(x+1)(x^2-9)} = \frac{A}{x+1} + \frac{B}{x-3} + \frac{C}{x+3}$, *find the value of* $C$

A. $\frac{1}{12}$ B. $-\frac{5}{4}$
C. $-\frac{13}{4}$ D. $\frac{13}{6}$

***[2021/22 e-Test]***

15. *If* $X = \{2, 4, 6\}$, *determine the number of power set of* $X$ *denoted as* $n(P(X))$

A. 7 B. 9 C. 8
D. 10 E. 11

***[2021/22 e-Test]***

16. *Given the universal set* $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ *where* $A = \{1, 3, 5, 7\}, B = \{1, 2, 5, 7\}$ *and* $C = \{3, 6, 7, 8\}$.
*List the members of* $A^c \cap B \cap C^c$

A. {1} B. {2} C. {3}
D. {4} E. {5}

***[2021/22 e-Test]***

17. *Given* $Z_1 = (3 + 2i)$ *and* $Z_2 = (4 - 5i)$, *evaluate* $Z_1 Z_2$

A. $2 - 7i$ B. $22 - 7i$
C. $2 + 7i$ D. $22 + 7i$

***[2021/22 e-Test]***

18. *Given that* $U = \{Natural\ numbers\}$,
$F = \{Factors\ of\ 20\}$ *and*
$M = \{Multiples\ of\ 3\ less\ than\ or\ equal\ to\ 15\}$.
*Find* $F \cap M$

A. {4, 5} B. {3, 5}
C. {2, 4} D. ∅

***[2021/22 e-Test]***

19. Evaluate $\sum_{r=1}^{4} r(r^2 + 1)$

A. 100 B. 110
C. 120 D. 130

***[2021/22 e-Test]***

20. In a class of 100 students, 40 offer chemistry, 35 offer geography and 30 offer neither. How many students offer both courses.

A. 5 B. 6
C. 15 D. 20

***[2021/22 e-Test]***

21. *Given set* $X = \{1, 2, 3\}$ *and set* $Y = \{3, 1, 2\}$, *which of the following statement is true for* $X$ *and* $Y$

A. $X \neq Y$ B. $X \leq Y$
C. $X \geq Y$ D. $X = Y$
E. $X \equiv Y$

***[2021/22 e-Test]***

22. *Given the universal set* $U = \{1, 2, 3, 4, 5\}$ *and* $P = \{1, 2, 4\}, Q = \{2, 4, 5\}$, *find* $P' \cap Q$

A. {1} B. {2} C. {3}
D. {4} E. {5}

***[2021/22 e-Test]***

23. $\frac{16}{9}, x, 1, y$ are G.P. Find the product of x and y.

A. 1 B. $\frac{1}{9}$
C. $-\frac{2}{3}$ D. 3

***[2021/22 e-Test]***

24. Express $\frac{(n+2)!}{n!}$

A. $(n + 1)(n + 2)$
B. $\frac{1}{2}(n + 1)(n + 2)$
C. $(n - 1)(n - 2)$
D. $n(n - 1)(n - 2)$

***[2021/22 e-Test]***

25. Obtain the sum of positive even integers less than 100.

A. 4950 B. 5050
C. 5500 D. 2450

***[2021/22 e-Test]***

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| ANSWERS | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | C | 6. | C | 11. | D | 16. | B | 21. | D |
| 2. | B | 7. | B | 12. | D | 17. | B | 22. | E |
| 3. | A | 8. | A | 13. | B | 18. | D | 23. | A |
| 4. | C | 9. | C | 14. | D | 19. | B | 24. | A |
| 5. | D | 10. | D | 15. | C | 20. | A | 25. | D |

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1) $z = 4 - 2\sqrt{2}i \qquad \bar{z} = 4 + 2\sqrt{2}i$

$z\bar{z} = (4 - 2\sqrt{2}i)(4 - 2\sqrt{2}i)$

$= (4)^2 + (2\sqrt{2})^2 = 16 + 8$

$= 24$ **C**

2.) $U = \{1, 2, 3, 4, 5, 6\}$

$X = \{2, 4, 6\}$

$Y = \{1, 2, 6\}$

$(X \cap Y) = \{2, 6\}$

$(X \cap Y)^c = \{1, 3, 4, 5\}$ **B**

3) $\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2 = \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$

$= \left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}i\right) + \left(\frac{\sqrt{3}}{2}i\right)^2$

$= \frac{1}{4} - \frac{\sqrt{3}}{2}i - \frac{3}{4} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

The imaginary part is $-\frac{\sqrt{3}}{2}$ **A**

4) $\frac{3x+2}{(x+2)(x-5)} = \frac{A}{x+2} + \frac{B}{x-5}$

$3x + 2 = A(x-5) + B(x+2)$

when $x = -2$

$3(-2) + 2 = A(-2-5) + B(-2+2)$

$-6 + 2 = -4 = A(-7) + 0$

$-7A = -4 \qquad A = 4/7$

when $x = 5$

$3(5) + 2 = A(5-5) + B(5+2)$

$15 + 2 = 17 = 0 + B(7)$

$7B = 17 \qquad B = 17/7$

Hence

$\frac{3x+2}{(x+2)(x-5)} = \frac{4}{7(x+2)} + \frac{17}{7(x-5)}$

$= \frac{1}{7}\left[\frac{4}{(x+2)} + \frac{17}{(x-5)}\right]$ **C**

5.) $z_1 = 2 - 4i \qquad z_2 = 6 + 7i$

$\frac{z_1}{z_2} = \frac{2-4i}{6+7i} = \frac{2-4i}{6+7i} \times \frac{6-7i}{6-7i}$

$= \frac{12 - 14i - 24i + 28i^2}{6^2 + 7^2}$

$= \frac{12 - 38i - 28}{36 + 49} = \frac{-16 - 38i}{85}$

$= -\left(\frac{16}{85} + \frac{36}{85}i\right)$ **D**

6.) Students that passed MATH111, $n(M) = 18$

Students that passed PHY113, $n(P) = 17$

Students that passed both, $n(M \cap P) = 11$

Students that passed MATH111 only:

$n(M \cap P') = 18 - 11 = 7$ **C**

7.) $z_1 = 1 - 3i \qquad z_2 = -2 + 5i$

$z_1 z_2 = (1 - 3i)(-2 + 5i)$

$= -2 + 5i + 6i - 15i^2$

$= -2 + 11i + 15 = 13 + 11i$

$z_1 + z_2 = (1 - 3i) + (-2 + 5i)$

$= -1 + 2i$

$\frac{z_1 z_2}{z_1 + z_2} = \frac{13 + 11i}{-1 + 2i}$

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$$\frac{z_1 z_2}{z_1 + z_2} = \frac{13+11i}{-1+2i} \times \frac{-1-2i}{-1+2i}$$

$$= \frac{-13 - 11i - 26i - 22i^2}{(-1)^2 + (2)^2}$$

$$= \frac{-13 - 37i + 22}{1+4} = \frac{9-37i}{5}$$

$$= \frac{9}{5} - \frac{37}{5}i$$

**B**

8. $$\frac{2+3i}{3+2i} = \frac{2+3i}{3+2i} \times \frac{3-2i}{3-2i}$$

$$= \frac{6 - 4i + 9i - 6i^2}{3^2 + 2^2}$$

$$= \frac{6 + 5i + 6}{9+4} = \frac{12+5i}{13}$$

The real part is $\frac{12}{13}$

**A**

9. $$\frac{3x^2 - 3x - 2}{(x-1)(x+1)} = A + \frac{Bx+C}{(x-1)(x+1)}$$

Using long division method

$(x-1)(x+1) = x^2 - 1$

$$\begin{array}{r} 3 \\ (x^2-1) \overline{)\ 3x^2 - 3x - 2} \\ -(3x^2 \quad -3) \\ \hline -3x + 1 \end{array}$$

Then

$$\frac{3x^2 - 3x - 2}{(x-1)(x+1)} = 3 + \frac{-3x+1}{(x-1)(x+1)}$$

Hence

$A = 3$, $B = -3$, $C = 1$

**D**

10. $$\frac{1}{(ax+b)^2} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2}$$

**D**

11. $$a + ib = \frac{1-i}{2+i}$$

$$\frac{1-i}{2+i} \times \frac{2-i}{2-i} = \frac{2 - i - 2i + i^2}{(2)^2 + (1)^2}$$

$$= \frac{2 - 3i - 1}{4+1} = \frac{1-3i}{5} = \frac{1}{5} - \frac{3}{5}i$$

Hence $a = \frac{1}{5}$

**D**

12. $$\frac{1}{ax^2 + bx + c}$$ (Given)

if $ax^2 + bx + c$ is factorisable into $(x+m)(x+n)$, we'll have

$$\frac{1}{ax^2 + bx + c} = \frac{A}{x+m} + \frac{B}{x+n}$$

if not factorisable

$$\frac{1}{ax^2 + bx + c} = \frac{Ax + B}{ax^2 + bx + c}$$

**D**

13. $$\frac{5+x}{(2x^2-3)(x-1)} = \frac{Ax+B}{2x^2-3} + \frac{C}{x-1}$$

$$5 + x = (Ax+B)(x-1) + C(2x^2-3)$$

when $x = 1$

$$5 + 1 = [A(1)+B](1-1) + C[2(1)^2 - 3]$$

$$6 = 0 + C(2-3)$$

$$6 = C(-1) = -C$$

$$C = -6$$

**B**

14. $\frac{x^2-4x+5}{(x+1)(x^2-9)} = \frac{A}{x+1} + \frac{B}{x-3} + \frac{C}{x+3}$

$x^2-4x+5 = A(x-3)(x+3) + B(x+1)(x+3) + C(x+1)(x-3)$

when $x = -3$

$(-3)^2 - 4(-3) + 5 = 0 + 0 + C[(-3+1)(-3-3)]$

$9 + 12 + 5 = C[(-2)(-6)] = 12C$

$26 = 12C, \quad C = \frac{26}{12} = \frac{13}{6}$ **D**

15) $X = \{2, 4, 6\}$ $n(X) = 3$

$n(P(X)) = 2^3 = 8$ **C**

which are : $\{2\}, \{4\}, \{6\}, \{2,4\}$ $\{2,6\}, \{4,6\}, \{2,4,6\}$ and $\emptyset$

16) $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 3, 5, 7\}$ $A^c = \{2, 4, 6, 8\}$

$B = \{1, 2, 5, 7\}$ $B^c = \{3, 4, 6, 8\}$

$C = \{3, 6, 7, 8\}$ $C^c = \{1, 2, 4, 5\}$

$A^c \cap B \cap C^c$

$= \{2, 4, 6, 8\} \cap \{1, 2, 5, 7\} \cap \{1, 2, 4, 5\}$

$= \{2\}$ **B**

17) $z_1 = 3 + 2i$ $z_2 = 4 - 5i$

$z_1 z_2 = (3+2i)(4-5i)$

$= 12 - 15i + 8i - 10i^2$

$= 12 - 7i + 10$

$= 22 - 7i$ **B**

18) $F = \{\text{factors of } 20\}$

$= \{1, 2, 4, 5, 10, 20\}$

$M = \{\text{multiple of } 3 \leq 15\}$

$= \{3, 6, 9, 15\}$

$F \cap M = \emptyset$ **D**

19. $\sum_{r=1}^{4} r(r^2+1)$

| r | $r^2$ | $r^2+1$ | $r(r^2+1)$ |
|---|---|---|---|
| 1 | 1 | 2 | 2 |
| 2 | 4 | 5 | 10 |
| 3 | 9 | 10 | 30 |
| 4 | 16 | 17 | 68 |
| | | | 110 |

**B**

20) Total number of students : $n(U) = 100$

Students that offer chemistry : $n(C) = 40$

Students that offer geography : $n(G) = 35$

students that offer neither : $n(C \cup G)' = 30$

students that offer both : $n(C \cap G) = x$

$n(U) = n(C) + n(G) - n(C \cap G) + n(C \cup G)'$

$100 = 40 + 35 - x + 30$

$100 = 105 - x$

$x = 5$ **A**

21) $X = \{1, 2, 3\}$ $Y = \{3, 1, 2\}$

Since the contain same element $X = Y$

* Two sets are equal (=) if thay contain same element
* Two sets are equivalent (≡) if they have the same cardinality or the same no of elements



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22. $\mathcal{U} = \{1, 2, 3, 4, 5\}$

$P = \{1, 2, 4\}$ $\quad P' = \{3, 5\}$

$Q = \{2, 4, 5\}$

$P' \cap Q = \{3, 5\} \cap \{2, 4, 5\}$

$= \{5\}$ **E**

23. For a three consecutive terms of a G.P., the square of the middle term equals the product of the other two terms.

Let say, the three consecutive terms of the G.P. are: $a$, $ar$ and $ar^2$

i.e $T_{n-1} = a$

$T_n = ar$

$T_{n+1} = ar^2$

$T_{n-1} \times T_{n+1} = a \times ar^2$

$= a^2r^2 = (ar)^2$

$= T_n^2$

Now considering terms 16, x, 1, y...

$xy = 1^2$, $xy = 1$ **A**

24. $\frac{(n+2)!}{n!} = \frac{(n+2)(n+1)n!}{n!}$

$= (n+1)(n+2)$ **A**

25. Sum of positive numbers less than 100?

The series: $2 + 4 + 6 + \ldots + 96 + 98$

first term, $a = 2$

last term, $L = 98$

Common difference, $d = 2$

To find the number of terms n present in the series:

$T_n = L = a + (n-1)d$

$98 = 2 + (n-1)2$

$98 = 2 + 2n - 2 = 2n$

$n = 49$

The number of terms is 49

Hence, using $S_n = \frac{n}{2}(a + L)$

$S_{49} = \frac{49}{2}(2 + 98) = \frac{49}{2}(100)$

$S_{49} = 2450$

Sum of positive numbers less than 100 is 2450 **D**